

Chapter 5 – Conductors and Dielectrics

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
2	Formulate electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.	Q2, M1, M2, F	C5	2

Objectives:

- 1. Current, Current Density and Ohm's Law
- 2. Conductor Properties and Boundary Conditions
- 3. Method of Images
- 4. Dielectric Materials and Boundary Conditions

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1

Electric Current

Current:

Current is electric charges in motion.

Symbol and Units:

I , ampere [A]

Definition:

Current is defined as rate of movement of charge passing through a given reference point (or crossing a given reference plane) of one Coulomb per second.

$$I = \frac{dQ}{dt}, \quad [1A] = \left[1 \frac{C}{s} \right]$$

Different Forms of Current:

Depending on how I is produced, there are different kinds of current:

- Convection Current
- Conduction Current
- Displacement Current

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Current Density

Current Density:

In field theory, we are more interested in events occurring at a point rather than within a large region, so current density is defined. Current Density is a vector quantity.

Symbol and Units:

$\mathbf{J}$, amperes per square meter $\left[\frac{\text{A}}{\text{m}^2}\right]$

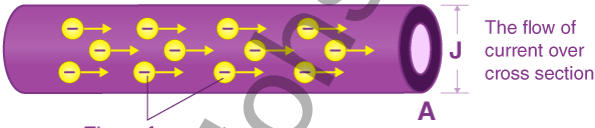
The increment of current ΔI crossing an incremental surface ΔS normal to the current density is, $\Delta I = J_N \Delta S$

and in the case where density is not perpendicular to the surface,

$$\Delta I = \mathbf{J} \cdot \Delta \mathbf{S}$$

Total current is obtained by integrating,

$$I = \int_S \mathbf{J} \cdot d\mathbf{S}$$



3

Relations of Current Density

Current Density as related to velocity of the moving charges:

Consider an element of charge, ΔQ

$$\Delta Q = \rho_v \Delta v$$
$$\Delta Q = \rho_v \Delta S \Delta L$$

In time interval Δt , the element of charge has moved a distance Δx through a ref plane perpendicular to the direction of motion,

$$\Delta Q = \rho_v \Delta S \Delta x$$

So, the resulting current is,

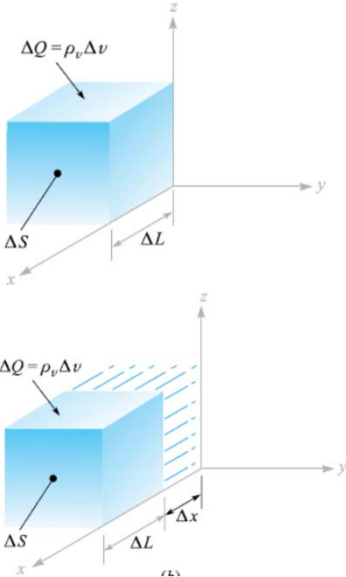
$$\Delta I = \frac{\Delta Q}{\Delta t} = \rho_v \Delta S \frac{\Delta x}{\Delta t}$$

The last expression then becomes,

$$\Delta I = \rho_v \Delta S v_x$$

where, v_x represents the x-comp of the velocity $\mathbf{v}$,
In terms of Current Density,

$$J_x = \rho_v v_x$$



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Relations of Current Density

Current Density as related to velocity of the moving charges:
In terms of Current Density,

$$J_x = \rho_v v_x$$

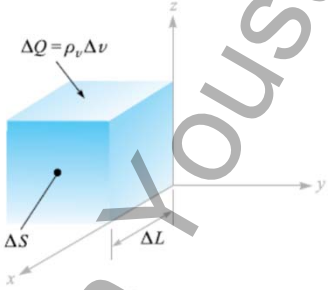
Similarly, for the element of charge moving along y and z-axes,

$$J_y = \rho_v v_y$$
$$J_z = \rho_v v_z$$

In vector form,

$$\mathbf{J} = \rho_v \mathbf{v}$$

Convection Current Density



$$\mathbf{J} = J_x \mathbf{a}_x + J_y \mathbf{a}_y + J_z \mathbf{a}_z$$
$$\mathbf{v} = v_x \mathbf{a}_x + v_y \mathbf{a}_y + v_z \mathbf{a}_z$$

Observation:

- Convection current density is related linearly to charge density as well as to velocity.
- Convection current different from conduction current does not involve conductors, and hence does not satisfy Ohm's Law.

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D5.1. Given the vector current density $\mathbf{J} = 10\rho^2 z \mathbf{a}_\rho - 4\rho \cos^2 \phi \mathbf{a}_\phi$ mA/m²:
(a) find the current density at $P(\rho = 3, \phi = 30^\circ, z = 2)$; (b) determine the total current flowing outward through the circular band $\rho = 3, 0 < \phi < 2\pi, 2 < z < 2.8$.

Given the vector density J, calculate the current density at P

$$\rho := 3 \qquad \phi := 30 \cdot \frac{\pi}{180} \qquad z := 2$$
$$\mathbf{J}(\rho, \phi, z) := \begin{pmatrix} 10 \cdot \rho^2 \cdot z \\ -4 \cdot \rho \cdot \cos^2(\phi) \\ 0 \end{pmatrix} \qquad \mathbf{J}(\rho, \phi, z) = \begin{pmatrix} 180 \\ -9 \\ 0 \end{pmatrix}$$

Determine the total current flowing outward through the circular band

$$\rho := 3 \qquad 0 < \phi < 2 \cdot \pi \qquad 2 < z < 2.8$$
$$I := \int_2^{2.8} \int_0^{2\pi} \mathbf{J}(\rho, \phi, z) \cdot \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} d\phi dz \qquad I = 3.257 \times 10^3$$

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Continuity of Current

Using Principal of Conservation of Charges,
Total current through a closed surface,

$$I = \oint_S \mathbf{J} \cdot d\mathbf{S}$$

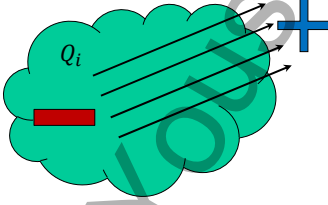
This outward flow of the positive charge must be balanced by a decrease of positive charge within the closed surface.
If the charge is Q_i within the closed surface, then the rate of decrease is,

$$I = -\frac{dQ_i}{dt}$$

And the principle of conservation of charge becomes,

$$I = \oint_S \mathbf{J} \cdot d\mathbf{S} = -\frac{dQ_i}{dt}$$

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Observation:
The presence or absence of a negative sign depends on what current and charge we consider. In circuits, usually conventional current is considered, i.e. due to positive charge. The negative sign in the expression is due to outward flow of this current.

Continuity of Current

Integral Form of Continuity Equation:

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = -\frac{dQ_i}{dt}$$

Point Form of Continuity Equation,
Using Divergence Theorem,

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \int_{vol} \nabla \cdot \mathbf{J} dv$$

Enclosed charge Q_i can be represented in terms of ρ_v ,

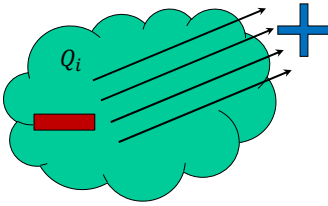
$$\int_{vol} \nabla \cdot \mathbf{J} dv = -\frac{d}{dt} \int_{vol} \rho_v dv$$

For constant surface,

$$\int_{vol} \nabla \cdot \mathbf{J} dv = \int_{vol} -\frac{\partial \rho_v}{\partial t} dv$$

Comparing the integrand on both sides,

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$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

Physical Interpretation:
This equation indicates that the current, or charge per second, diverging from a small volume per unit volume is equal to the time rate of decrease of charge per unit volume at every point.

Point Form of Continuity of Current

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

Numerical Example:

For the given current density, calculate the outward current at (a) $r = 5m$ and (b) $r = 6m$ at an instant of time elapsed, $t = 1$ sec.

Current Density is given as,

$$\mathbf{J} = \frac{1}{r} e^{-t} \mathbf{a}_r, \quad [\text{A/m}^2]$$

Current can be found from the integral expression as,

$$I = \int_S \mathbf{J} \cdot d\mathbf{S}_r = \int_{\varphi} \int_{\theta} \frac{1}{r} e^{-t} r^2 \sin \theta d\theta d\varphi$$

$$I = r e^{-t} \int_{\varphi=0}^{2\pi} \int_{\theta=0}^{\pi} \sin \theta d\theta d\varphi = 4\pi r e^{-t}$$

$$I_{5m} = 4\pi 5 e^{-1} = 23.1 \text{ A}$$

$$I_{6m} = 4\pi 6 e^{-1} = 27.7 \text{ A}$$

Thus, the total current is larger at $r = 6m$ than it is at $r = 5m$. Why?

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Point Form of Continuity of Current

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

Numerical Example:

Solution for Why?

$$\mathbf{J} = \frac{1}{r} e^{-t} \mathbf{a}_r$$

Using continuity equation,

$$-\frac{\partial \rho_v}{\partial t} = \nabla \cdot \mathbf{J} = \nabla \cdot \left(\frac{1}{r} e^{-t} \mathbf{a}_r \right)$$

$$-\frac{\partial \rho_v}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{1}{r} e^{-t} \right) = \frac{1}{r^2} e^{-t}$$

Finding ρ_v by integrating w.r.t. t ,

$$\rho_v = -\int \frac{1}{r^2} e^{-t} dt + K(r) = \frac{1}{r^2} e^{-t} + K(r)$$

The velocity is greater at $r = 6$ than it is at $r = 5$, as some force is accelerating the charge density in an outward direction

$$\text{div } \mathbf{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r)$$

If we assume that $\rho_v \rightarrow 0$ as $t \rightarrow \infty$, then $K(r) = 0$, and

$$\rho_v = \frac{1}{r^2} e^{-t}, \quad [\text{C/m}^3]$$

Now,

$$\mathbf{J} = \rho_v \mathbf{v}$$

So,

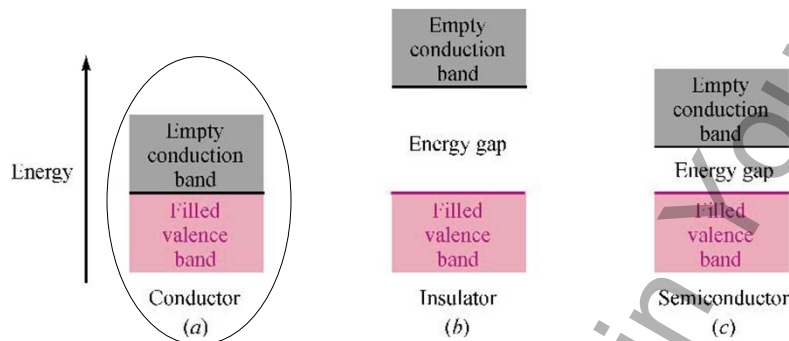
$$\mathbf{v} = \frac{\mathbf{J}}{\rho_v} = \frac{\frac{1}{r} e^{-t}}{\frac{1}{r^2} e^{-t}} = r \left[\frac{m}{s} \right]$$

D5.2.

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Conductors, Dielectrics, Semiconductors



The Energy Band Structure in Three Different Types of Materials at 0K

- The conductor exhibits no energy gap between the valence and conduction bands.
- The Insulator shows a large energy gap
- The semiconductor has only a small energy gap

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Conduction Current in Metallic Conductors

- In a conductor, valence electrons, or *conduction*, or *free*, electrons, move under the influence of an electric field, $\mathbf{E}$.
- Force experienced by an electron having a charge, $Q = -e$, is

$$\mathbf{F} = Q\mathbf{E} = -e\mathbf{E}$$
- In free space, the electron would accelerate and continuously increase its velocity (and energy); in the crystalline material, the progress of the electron is impeded by continual collisions with the thermally excited crystalline lattice structure, and a constant average velocity is soon attained, known as *drift velocity*, $\mathbf{v}_d$.
- Drift velocity is linearly related to $\mathbf{E}$ by the *mobility* of the electron in the given material,

$$\mathbf{v}_d = -\mu_e \mathbf{E}$$

where μ_e is the mobility of an electron and is positive by definition.

- This electron velocity is in a direction opposite to the direction of $\mathbf{E}$.

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Conduction Current in Metallic Conductors

- Current Density is,

$$\mathbf{J} = \rho_v \mathbf{v}_d, \quad \mathbf{v}_d = -\mu_e \mathbf{E}$$
where ρ_v contains both electrons and holes,

$$\rho_v = \rho_e + \rho_h$$
where ρ_e is the free-electron charge density, a negative value,

$$\mathbf{J} = -\rho_e \mu_e \mathbf{E}, \quad [1]$$
- So, the current density and electric field intensity are in the same direction.
- $\mathbf{J}$ and $\mathbf{E}$ are related by conductivity σ (sigma) in metallic conductors,

$$\mathbf{J} = \sigma \mathbf{E}, \quad [2]$$
- Conductivity is measured in Siemens [S], $1\text{S} = 1 \frac{\text{A}}{\text{V}} = \text{mhos} = \frac{1}{\Omega} = \Omega^{-1}$

$$\sigma = \frac{1}{\rho}, \quad [\text{reciprocal of Resistivity}]$$
- Comparing eq [1] and [2],

$$\sigma = -\rho_e \mu_e$$

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Conduction Current in Metallic Conductors

- Assume $\mathbf{J}$ and $\mathbf{E}$ are uniform in the cylindrical region shown.

Conductivity σ

Area = S

I

E

L

$I = \int_S \mathbf{J} \cdot d\mathbf{S} = JS$

$V_{ab} = -\int_b^a \mathbf{E} \cdot d\mathbf{L} = -E \cdot \int_b^a dL$

$V_{ab} = -\mathbf{E} \cdot \mathbf{L}_{ba}$

$V_{ab} = \mathbf{E} \cdot \mathbf{L}_{ab}$

$V = EL$

$R = \frac{V}{I}$

$R = \frac{V_{ab}}{I} = \frac{-\int_b^a \mathbf{E} \cdot d\mathbf{L}}{\int_S \mathbf{J} \cdot d\mathbf{S}}$

$\mathbf{J} = \sigma \mathbf{E}$

Point form of Ohm's Law

$$J = \frac{I}{S} = \sigma \frac{V}{L}$$
$$V = \frac{L}{\sigma S} I = \frac{\rho L}{S} I$$
$$V = RI$$
$$R = \frac{L}{\sigma S} = \frac{\rho L}{S}$$

D5.3. D5.4.

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Conductor Properties and Boundary Conditions

- Assume free electrons show up within a perfect conductor. The result will be an electric field due to negative charged particles (electrons) not counteracted by positive charges and they move randomly reaching the surface but can't exit from the surface of the conductor as bounded by an insulator (outside the surface). So, as a consequence, no charge may remain within the conductor. If it did, the resulting electric field would force the charge to the surface. Hence, charge density ρ_v is zero within a conductor and all the charges reside at the surface of the conductor.
- For static conditions, no current may flow, i.e. $E = 0$ within the conductor, follows directly from Ohm's Law, because if $E \neq 0$, the conduction electrons would move and produce a current, thus leading to a non-static condition.
- Summarizing for electrostatics, *no charge* and *no electric field* may exist at any point *within* a conducting material. Charge, may however, appear on the surface as a **surface charge density**, requiring to investigate the field external to the surface.

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Conductor Properties and Boundary Conditions

Qualitative Analysis of Fields external to the Conductor

- External electric field intensity is decomposed into two components, one *tangential* and one *normal* to the conducting surface.
- The tangential component is zero, because if it were not zero, a tangential force would be applied to the surface charge, resulting in their motion and cause non-static conditions against our original assumption (**as it will cause a current to flow without any external field**).
- For normal component, the **charge residing at the surface of the conductor** would be equal to the normal component of the flux density there pointing outward from the surface as it can't penetrate inside, for the total field there is zero.

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Conductor Properties and Boundary Conditions

Quantitative Analysis of Fields external to the Conductor

➤ Let us set up a boundary between a conductor and free space showing tangential and normal components of **D** and **E** on the free-space side of the boundary.

Tangential Field
KVL is applied
 $\oint \mathbf{E} \cdot d\mathbf{L} = 0$

For the closed path *abcd*, $\int_a^b + \int_b^c + \int_c^d + \int_d^a = 0$

$$E_t \Delta w - E_{N,at\,b} \frac{1}{2} \Delta h + E_{N,at\,a} \frac{1}{2} \Delta h = 0$$

Free Space

Conductor

Using,
 $\mathbf{D} = \epsilon_0 \mathbf{E}$
 $D_t = 0$

$E_t \Delta w = 0$
 $E_t = 0$

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Conductor Properties and Boundary Conditions

Quantitative Analysis of Fields external to the Conductor

➤ Let us set up a boundary between a conductor and free space showing tangential and normal components of **D** and **E** on the free-space side of the boundary.

Normal Field
Gauss's Law is applied
 $\oint \mathbf{D} \cdot d\mathbf{S} = Q_{encl}$

For the three surfaces, $\int_{top} + \int_{bottom} + \int_{sides} = Q$

$$D_N \Delta S + 0 + 0 = Q_{encl}$$

Free Space

Conductor

Using,
 $\mathbf{D} = \epsilon_0 \mathbf{E}$
 $E_N = \frac{\rho_s}{\epsilon_0}$

$D_N \Delta S = \rho_s \Delta S$
 $D_N = \rho_s$

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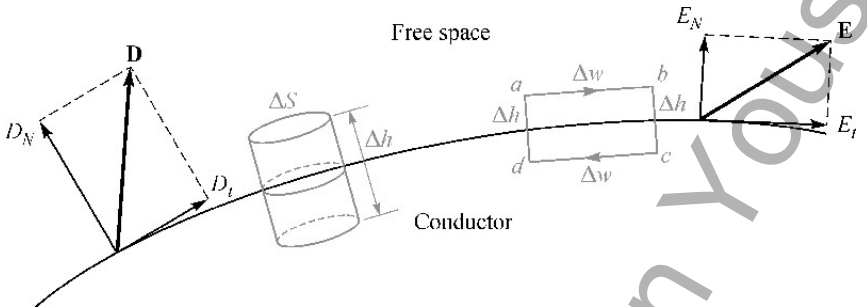
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Conductor Properties and Boundary Conditions

Boundary Conditions for the Conductor-to-free-space boundary



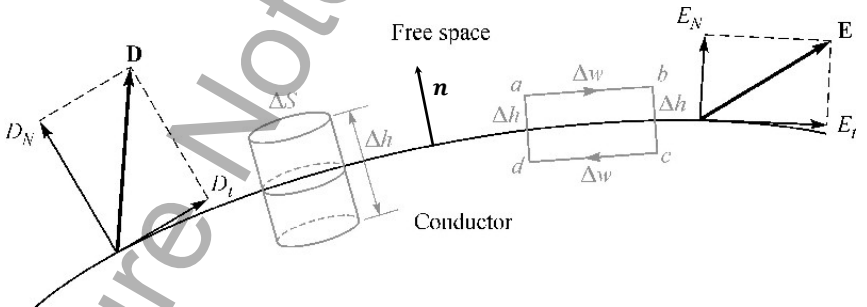
Tangential Field (KVL)
$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$
$$D_t = E_t = 0$$

Normal Field (Gauss's Law)
$$\oint \mathbf{D} \cdot d\mathbf{S} = Q_{encl}$$
$$D_N = \epsilon_0 E_N = \rho_S$$

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Conductor Properties and Boundary Conditions

Boundary Conditions for the Conductor-to-free-space boundary



$$D_t = E_t = 0$$
$$D_N = \epsilon_0 E_N = \rho_S$$

Tangential Component
$$\mathbf{E} \times \mathbf{n} \Big|_S = 0$$

Normal Component
$$\mathbf{D} \cdot \mathbf{n} \Big|_S = \rho_S$$

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Conductor Properties and Boundary Conditions

Properties of a Good Conductor

- 1) The static electric field intensity inside a conductor is zero.
- 2) The static electric field at the surface of a conductor is everywhere directed normal to that surface.
- 3) The conductor surface is an equipotential surface.

(Zero tangential electric field, $E_t = 0$)

EXAMPLE 5.2

Given the potential,

$$V = 100(x^2 - y^2)$$

and a point $P(2, -1, 3)$ that is stipulated to lie on a conductor-to-free-space boundary, find V , $\mathbf{E}$, $\mathbf{D}$, and ρ_s at P , and also the equation of the conductor surface.

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$$V_P = 100[2^2 - (-1)^2] = 300 \text{ V}$$

$$V = 100(x^2 - y^2)$$

$$300 = 100(x^2 - y^2)$$

$$x^2 - y^2 = 3$$

$$\mathbf{E} = -100\nabla(x^2 - y^2) = -200x\mathbf{a}_x + 200y\mathbf{a}_y$$

$$\mathbf{E}_P = -400\mathbf{a}_x - 200\mathbf{a}_y \text{ V/m}$$

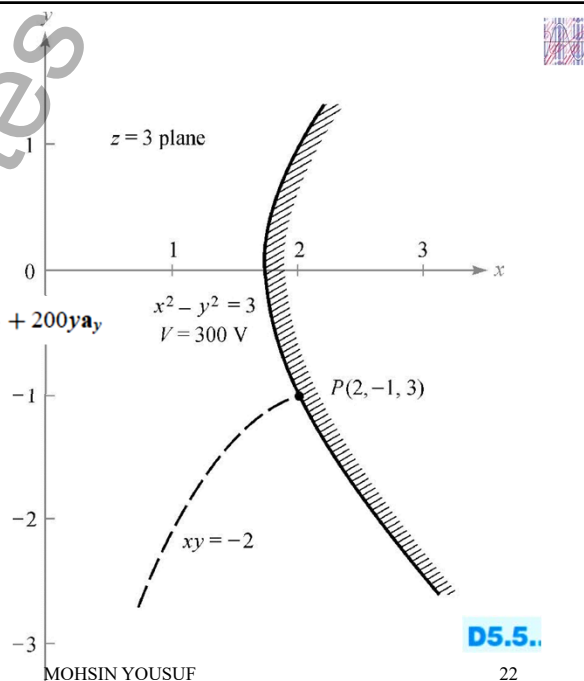
$$\mathbf{D} = \epsilon_0 \mathbf{E}$$

$$\mathbf{D}_P = 8.854 \times 10^{-12} \mathbf{E}_P$$

$$= -3.54\mathbf{a}_x - 1.771\mathbf{a}_y \text{ nC/m}^2$$

$$D_N = |\mathbf{D}_P| = 3.96 \text{ nC/m}^2$$

$$\rho_{s,P} = D_N = 3.96 \text{ nC/m}^2$$



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The Method of Images

Equipotential surface, $V=0$

(a)

Conducting plane, $V=0$

(b)

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The Method of Images

Conducting plane, $V=0$

(a)

Equipotential surface, $V=0$

(b)

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The Method of Images

(a)

(b)

$$\mathbf{E}_+ = \frac{\rho_L}{2\pi\epsilon_0 R_+} \mathbf{a}_{R+} = \frac{30 \times 10^{-9}}{2\pi\epsilon_0 \sqrt{13}} \frac{2\mathbf{a}_x - 3\mathbf{a}_z}{\sqrt{13}}$$
$$\mathbf{E}_- = \frac{30 \times 10^{-9}}{2\pi\epsilon_0 \sqrt{13}} \frac{2\mathbf{a}_x + 3\mathbf{a}_z}{\sqrt{13}}$$
$$\mathbf{E} = \frac{-180 \times 10^{-9} \mathbf{a}_z}{2\pi\epsilon_0 (13)} = -249 \mathbf{a}_z \text{ V/m}$$
$$\mathbf{D} = \epsilon_0 \mathbf{E} = -2.20 \mathbf{a}_z \text{ nC/m}^2$$

D5.6.

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The Nature of Dielectric Materials

Most solid materials are classified as insulators because they offer very large resistance to the flow of electric current. Metals are classified as conductors because their outer electrons are not tightly bound, but in most materials even the outermost electrons are so tightly bound that there is essentially zero electron flow through them with ordinary voltages. Some materials are particularly good insulators and can be characterized by their high resistivities:

	Resistivity (ohm)
Glass	10^{12}
Mica	9×10^{13}
Quartz (fused)	5×10^{16}

	Resistivity (ohm)
Copper	1.7×10^{-8}

-Dielectric in an electric field can be viewed as a free-space arrangement of microscopic electric dipoles which are composed of positive and negative charges whose centers do not quite coincide.

- Not free charges – They are bound charges
- They are sources of electrostatic fields

Model – Polarization **P** and Permittivity

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The Nature of Dielectric Materials

- ❑ If a material contains **polar molecules**, they will generally be in random orientations when no electric field is applied. An applied electric field will polarize the material by orienting the dipole moments of polar molecules. This decreases the effective electric field between the plates and will increase the capacitance of the parallel plate structure. *The dielectric must be a good electric insulator so as to minimize any DC leakage current through a capacitor.*
- ❑ Polar molecules have a permanent displacement existing between the centers of gravity of the positive and negative charges, and each pair of charges acts as a dipole. Dipoles are oriented randomly.
- ❑ A **non-polar molecule** does not have this dipole arrangement until a field is applied.
- ❑ A dipole may be described by its dipole moment p
- ❑ $p = Qd$ where d is the vector from the negative to the positive charge.
 p in coulomb-meters [C m]

$$p_{total} = \sum_{i=1}^{n \Delta v} p_i$$

- ❑ Polarization is defined as **dipole moment per unit volume**.

$$P = \lim_{\Delta v \rightarrow 0} \frac{1}{\Delta v} \sum_{i=1}^{n \Delta v} p_i$$

The Nature of Dielectric Materials

Electrical Properties	ASTM Standard	Unit	Teflon® PTFE Powder, Paste, Disper.	Teflon® FEP	Teflon® PFA	Tefzel®
Dielectric Constant	D150	1 MHz	2.1	2.1	2.1	2.6
Dissipation Factor	D150	1MHz	<0.0001	0.0006	0.0001	0.007
Arc Resistance	D495	sec	>300	>300	>180	122
Volume Resistivity	D257	ohm·cm	>10 ¹⁸	>10 ¹⁸	>10 ¹⁸	>10 ¹⁷
Surface Resistivity	D257	ohm·sq	>10 ¹⁸	>10 ¹⁶	>10 ¹⁷	>10 ¹⁵

The Nature of Dielectric Materials

Electrical Properties of Kapton®
Type HN Polyimide Film

Property	0.30	0.50	1.00	2.00	3.00	5.00
	(7.6)	(12.7)*	(25.4)*	(50.8)*	(76.2)*	(127)*
Dielectric Strength, AC V/mil (kV/mm), min.	3,000 (118)	3,000 (118)	6,000 (236)	5,000 (197)	4,500 (177)	3,000 (118)
Volume Resistivity, ohm-cm at 200°C (392°F), min.	10 ¹²	10 ¹²	10 ¹²	10 ¹²	10 ¹²	10 ¹²
Dielectric Constant at 1 kHz, max.	4.0	4.0	3.9	3.9	3.9	3.9
Dissipation Factor at 1 kHz, max.	0.0070	0.0050	0.0036	0.0036	0.0036	0.0036

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The Nature of Dielectric Materials

Net total charge which crosses the surface in an upward direction,
 $\Delta Q_b = nQd \cdot \Delta S$
 $\Delta Q_b = \mathbf{P} \cdot \Delta \mathbf{S}$
 $Q_b = - \oint_S \mathbf{P} \cdot d\mathbf{S}$
Total enclosed charges, bound plus free:
 $Q_T = \oint_S \epsilon_0 \mathbf{E} \cdot d\mathbf{S}$
where,
 $Q_T = Q_b + Q$
 $Q = Q_T - Q_b = \text{free charge}$
 $Q = \oint_S \epsilon_0 \mathbf{E} \cdot d\mathbf{S} + \oint_S \mathbf{P} \cdot d\mathbf{S}$
 $Q = \oint_S (\epsilon_0 \mathbf{E} + \mathbf{P}) \cdot d\mathbf{S} = \oint_S \mathbf{D} \cdot d\mathbf{S}$
 $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$

Dielectric material

$\Delta \mathbf{S}$
 $\mathbf{E}$

Charge Densities

$Q_b = \int_v \rho_b dv$
 $Q = \int_v \rho_v dv$
 $Q_T = \int_v \rho_T dv$

Divergence

$\nabla \cdot \mathbf{P} = -\rho_b$
 $\nabla \cdot \epsilon_0 \mathbf{E} = \rho_T$
 $\nabla \cdot \mathbf{D} = \rho_v$

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The Nature of Dielectric Materials

The linear relationship between $\mathbf{P}$ and $\mathbf{E}$ is,

$$\mathbf{P} = \chi_e \epsilon_0 \mathbf{E}$$

where, χ_e (chi) is a dimensionless quantity called the **electric susceptibility** of the material.

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$
$$\mathbf{D} = \epsilon_0 \mathbf{E} + \chi_e \epsilon_0 \mathbf{E}$$
$$\mathbf{D} = (1 + \chi_e) \epsilon_0 \mathbf{E}$$

where,

$$\epsilon_r = (1 + \chi_e)$$

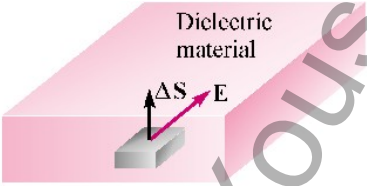
is known as **relative permittivity**, or dielectric constant of the material.

$$\mathbf{D} = \epsilon_0 \epsilon_r \mathbf{E} = \epsilon \mathbf{E}$$

where,

$$\epsilon = \epsilon_0 \epsilon_r$$

is the **permittivity**.



Dielectric material

Important Relationships

$$\mathbf{D} = \epsilon \mathbf{E}$$
$$\epsilon = \epsilon_0 \epsilon_r$$

Gauss's Law

$$\nabla \cdot \mathbf{D} = \rho_v$$
$$Q = \oint_S \mathbf{D} \cdot d\mathbf{S} = Q_T - Q_b$$

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The Nature of Dielectric Materials

EXAMPLE 5.4

We locate a slab of Teflon in the region $0 \leq x \leq a$, and assume free space where $x < 0$ and $x > a$. Outside the Teflon there is a uniform field $\mathbf{E}_{\text{out}} = E_0 \mathbf{a}_x$ V/m. We seek values for $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{P}$ everywhere.

Solution. The dielectric constant of the Teflon is 2.1, and thus the electric susceptibility is 1.1.

Outside the Slab $\mathbf{D}_{\text{out}} = \epsilon_0 E_0 \mathbf{a}_x$ $\mathbf{P}_{\text{out}} = 0$.

Inside the Slab

$$\mathbf{D}_{\text{in}} = 2.1 \epsilon_0 \mathbf{E}_{\text{in}} \quad (0 \leq x \leq a)$$
$$\mathbf{P}_{\text{in}} = 1.1 \epsilon_0 \mathbf{E}_{\text{in}} \quad (0 \leq x \leq a)$$

D5.8. A slab of dielectric material has a relative dielectric constant of 3.8 and contains a uniform electric flux density of 8 nC/m^2 . If the material is lossless, find: (a) E ; (b) P ; (c) the average number of dipoles per cubic meter if the average dipole moment is $10^{-29} \text{ C} \cdot \text{m}$.

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Boundary Conditions For Perfect Dielectric Materials

$\oint \mathbf{E} \cdot d\mathbf{L} = 0$

Region 1 ϵ_1

Region 2 ϵ_2

$\mathbf{n}$

D_{N1}

D_{N2}

ΔS

$E_{tan\ 1}$

$E_{tan\ 2}$

Δw

Δh

$E_{tan\ 1} \Delta w - E_{tan\ 2} \Delta w = 0$

$\frac{D_{tan\ 1}}{\epsilon_1} = E_{tan\ 1} = E_{tan\ 2} = \frac{D_{tan\ 2}}{\epsilon_2}$

$\frac{D_{tan\ 1}}{D_{tan\ 2}} = \frac{\epsilon_1}{\epsilon_2}$

Continuous

Discontinuous

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Boundary Conditions For Perfect Dielectric Materials

$\oint \mathbf{E} \cdot d\mathbf{L} = 0$

Region 1 ϵ_1

Region 2 ϵ_2

$\mathbf{n}$

D_{N1}

D_{N2}

ΔS

$E_{tan\ 1}$

$E_{tan\ 2}$

Δw

Δh

$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{encl}$

$D_{N1} \Delta S - D_{N2} \Delta S = \Delta Q = \rho_s \Delta S$

$D_{N1} - D_{N2} = \rho_s$

$D_{N1} = D_{N2}$

Continuous

$(\mathbf{D}_1 - \mathbf{D}_2) \cdot \mathbf{n} = \rho_s$

$(\mathbf{E}_1 - \mathbf{E}_2) \times \mathbf{n} = 0$

$\epsilon_1 E_{N1} = \epsilon_2 E_{N2}$

Discontinuous

These boundary conditions can be used to show the change in the vectors $\mathbf{D}$ and $\mathbf{E}$ at the surface.

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Boundary Conditions For Perfect Dielectric Materials

Let $\mathbf{D}_1$ (and $\mathbf{E}_1$) make an angle θ_1 with a normal to the surface, find D_2 .

$\epsilon_1 > \epsilon_2$

$D_{N1} = D_{N2}$ Continuous

$D_1 \cos \theta_1 = D_2 \cos \theta_2, \quad (1)$

$\frac{D_{\tan 1}}{D_{\tan 2}} = \frac{\epsilon_1}{\epsilon_2}$ Discontinuous

$\frac{D_1 \sin \theta_1}{D_2 \sin \theta_2} = \frac{\epsilon_1}{\epsilon_2}, \quad (2)$

$\frac{eq(2)}{eq(1)} \Rightarrow \frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2}, \quad (3) \quad \theta_1 > \theta_2$

$eq(2) \Rightarrow \epsilon_2 D_1 \sin \theta_1 = \epsilon_1 D_2 \sin \theta_2, \quad (4)$

$D_2 \sin \theta_2 = \frac{\epsilon_2}{\epsilon_1} D_1 \sin \theta_1, \quad (5)$

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Boundary Conditions For Perfect Dielectric Materials

Let $\mathbf{D}_1$ (and $\mathbf{E}_1$) make an angle θ_1 with a normal to the surface, find D_2 .

$\epsilon_1 > \epsilon_2$

$D_2 \sin \theta_2 = \frac{\epsilon_2}{\epsilon_1} D_1 \sin \theta_1, \quad (5)$

$eq(1)^2 \Rightarrow$

$D_1^2 \cos^2 \theta_1 = D_2^2 \cos^2 \theta_2$

$\because \cos^2 \theta + \sin^2 \theta = 1$

$D_2^2 (1 - \sin^2 \theta_2) = D_1^2 \cos^2 \theta_1$

$D_2^2 = D_1^2 \cos^2 \theta_1 + D_2^2 \sin^2 \theta_2$

$eq(5)^2 \Rightarrow D_2^2 = D_1^2 \cos^2 \theta_1 + \left(\frac{\epsilon_2}{\epsilon_1}\right)^2 D_1^2 \sin^2 \theta_1$

$D_2 = D_1 \sqrt{\cos^2 \theta_1 + \left(\frac{\epsilon_2}{\epsilon_1}\right)^2 \sin^2 \theta_1}$

$D_1 \cos \theta_1 = D_2 \cos \theta_2, \quad (1)$

$\frac{D_1 \sin \theta_1}{D_2 \sin \theta_2} = \frac{\epsilon_1}{\epsilon_2}, \quad (2)$

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Boundary Conditions For Perfect Dielectric Materials

EXAMPLE 5.5

We locate a slab of Teflon in the region $0 \leq x \leq a$, and assume free space where $x < 0$ and $x > a$. Outside the Teflon there is a uniform field $\mathbf{E}_{\text{out}} = E_0 \mathbf{a}_x$ V/m. We seek values for $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{P}$ everywhere.

Complete Example 5.4 by finding the fields within the Teflon ($\epsilon_r = 2.1$), given the uniform external field $\mathbf{E}_{\text{out}} = E_0 \mathbf{a}_x$ in free space.

Outside the Slab $\mathbf{D}_{\text{out}} = \epsilon_0 E_0 \mathbf{a}_x$ $\mathbf{P}_{\text{out}} = 0$.

Inside the Slab $\mathbf{D}_{\text{in}} = 2.1 \epsilon_0 \mathbf{E}_{\text{in}}$ ($0 \leq x \leq a$)

$D_{N1} = D_{N2}$

Continuous

$\mathbf{D}_{\text{in}} = \mathbf{D}_{\text{out}} = \epsilon_0 E_0 \mathbf{a}_x$

$\mathbf{D}_{\text{in}} = \epsilon \mathbf{E}_{\text{in}}$

$\mathbf{E}_{\text{in}} = \frac{\mathbf{D}_{\text{in}}}{\epsilon} = \frac{\epsilon_0 E_0 \mathbf{a}_x}{\epsilon_0 \epsilon_r} = 0.476 E_0 \mathbf{a}_x$

Summary

$\mathbf{P}_{\text{in}} = \mathbf{D}_{\text{in}} - \epsilon_0 \mathbf{E}_{\text{in}}$
 $\mathbf{P}_{\text{in}} = \epsilon_0 E_0 \mathbf{a}_x - 0.476 \epsilon_0 E_0 \mathbf{a}_x$
 $\mathbf{P}_{\text{in}} = 0.524 \epsilon_0 E_0 \mathbf{a}_x$

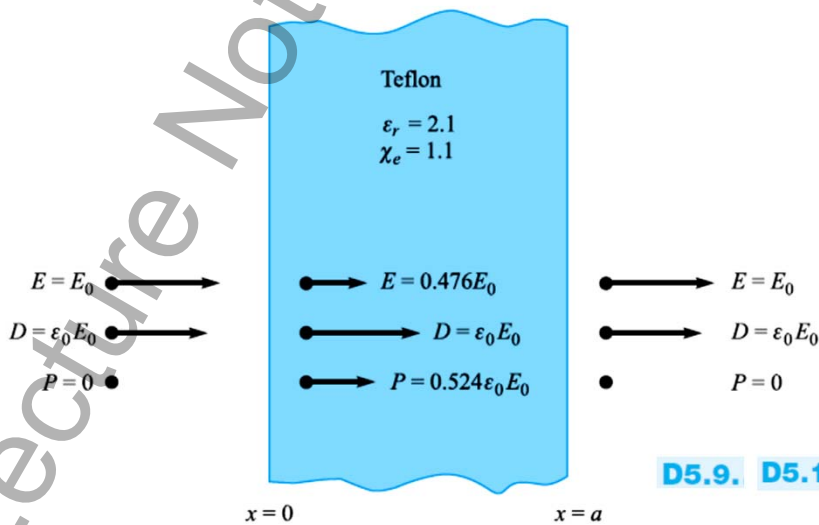
$\mathbf{D}_{\text{in}} = \epsilon_0 E_0 \mathbf{a}_x$ ($0 \leq x \leq a$)
 $\mathbf{E}_{\text{in}} = 0.476 E_0 \mathbf{a}_x$ ($0 \leq x \leq a$)
 $\mathbf{P}_{\text{in}} = 0.524 \epsilon_0 E_0 \mathbf{a}_x$ ($0 \leq x \leq a$)

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Boundary Conditions For Perfect Dielectric Materials

EXAMPLE 5.5



D5.9. D5.10.

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Boundary Conditions For Perfect Dielectric Materials

EXAMPLE 5.5

Teflon

$\epsilon_r = 2.1$

$\chi_e = 1.1$

$E = E_0$

$D = \epsilon_0 E_0$

$P = 0$

$E = 0.476 E_0$

$D = \epsilon_0 E_0$

$P = 0.524 \epsilon_0 E_0$

$E = E_0$

$D = \epsilon_0 E_0$

$P = 0$

$x = 0$

$x = a$

D5.9. D5.10.

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The Nature of Dielectric Materials

Electric Dipole

much more common than isolated

Charges

is characterized by

Dipole Moment

and

Electric Field of a Dipole

Potential of a Dipole

when placed in an

Electric Field

experiences

Torque on Dipole

and possesses

Energy of Dipole

is used to describe

Polarization of Dielectric

is important in

Molecular Dipole Moments

like that of the

Water Molecule

for application in

Capacitors

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